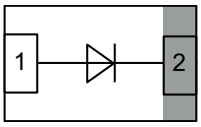
# Small Signal Fast Switching Diode






## FEATURES

- Silicon epitaxial planar diode
- Fast switching diode
- Leadless ultra small DFN1006-2A package (1 mm × 0.6 mm × 0.45 mm)
- Power dissipation better than SOT-23
- Surface-mounted device (SMD) plastic package with visible and sidewall plated / wettable flanks
- Soldering can be checked by standard visual inspection. No X-ray inspection necessary to meet automotive AOI requirements
- AEC-Q101 qualified available
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

AUTOMOTIVE GRADE Available

Pb-free

e3

RoHS COMPLIANT

HALOGEN FREE GREEN (5-2008)

## LINKS TO ADDITIONAL RESOURCES

3D Models | Models | Application Notes

## MECHANICAL DATA

**Case:** DFN1006-2A

**Weight:** 0.83 mg

**Molding compound flammability rating:** UL 94 V-0

**Terminals:** high temperature soldering guaranteed: Peak temperature max. 260 °C

**Packaging codes / options:**

08/10K per 7" reel (8 mm tape)

**PARTS TABLE**

| PART | ORDERING CODE | AEC-Q101 QUALIFIED | CIRCUIT CONFIGURATION | TYPE MARKING | REMARKS |
|---|---|---|---|---|---|
| BAS16L | BAS16L-G3-08 | no | Single | .D | Tape and reel |
| | BAS16L-HG3-08 | yes | | | |

**ABSOLUTE MAXIMUM RATINGS** ($T_{amb}$ = 25 °C, unless otherwise specified)

| PARAMETER | TEST CONDITION | SYMBOL | VALUE | UNIT |
|---|---|---|---|---|
| Reverse voltage | | $V_R$ | 100 | V |
| Forward current | on FR-4 board with recommended soldering footprint | $I_F$ | 250 | mA |
| Non repetitive forward current (1) | $t_p$ = 1 µs | $I_{FSM}$ | 9 | A |
| | $t_p$ = 1 ms | | 1.7 | |
| | $t_p$ = 1 s | | 1 | |
| Repetitive peak forward current | $T_L$ = 100 °C, $t_p$ = ≤ 1 ms, D = 0.05 | $I_{FRM}$ | 500 | mA |
| Power dissipation | on FR-4 board with recommended soldering footprint | $P_{tot}$ | 300 | mW |
| | $R_{thJL}$ = 100 K/W | | 1250 | mW |

**Note**

(1) Square wave, $T_j$ = 25 °C prior to surge

**THERMAL CHARACTERISTICS** ($T_{amb}$ = 25 °C, unless otherwise specified)

| PARAMETER | TEST CONDITION | SYMBOL | VALUE | UNIT |
|---|---|---|---|---|
| Thermal resistance junction to ambient air | according to JEDEC® 51-3 on FR-4 board with recommended soldering footprint | $R_{thJA}$ | 420 | K/W |
| Thermal resistance junction to lead | | $R_{thJL}$ | 100 | K/W |
| Maximum junction temperature | | $T_{j\ max.}$ | 150 | °C |
| Storage temperature range | | $T_{stg}$ | -55 to +150 | °C |
| Operating temperature range | | $T_{op}$ | -55 to +150 | °C |



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| **ELECTRICAL CHARACTERISTICS** ($T_{amb}$ = 25 °C, unless otherwise specified) | | | | | |
|---|---|---|---|---|---|
| **PARAMETER** | **TEST CONDITION** | **SYMBOL** | **TYP.** | **MAX.** | **UNIT** |
| Forward voltage | $I_F$ = 150 mA | $V_F$ | | 1.250 | V |
| | $I_F$ = 50 mA | | | 1.0 | V |
| | $I_F$ = 10 mA | | | 0.86 | V |
| | $I_F$ = 1 mA | | | 0.715 | V |
| Leakage current | $V_R$ = 80 V | $I_R$ | | 500 | nA |
| | $V_R$ = 80 V, $T_J$ = 150 °C | $I_R$ | | 100 | µA |
| | $V_R$ = 100 V | $I_R$ | | 1 | µA |
| Diode capacitance | $V_R$ = 0 V, f = 1 MHz | $C_D$ | 0.36 | 2 | pF |
| Reverse recovery time | $I_F$ = 10 mA, $I_R$ = 10 mA, $i_R$ = 1 mA | $t_{rr}$ | | 4 | ns |

## TYPICAL CHARACTERISTICS ($T_{amb}$ = 25 °C, unless otherwise specified)

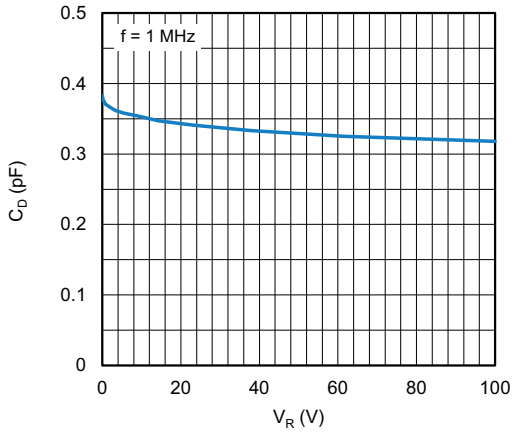



Fig. 1 - Typical Capacitance vs. Reverse Voltage

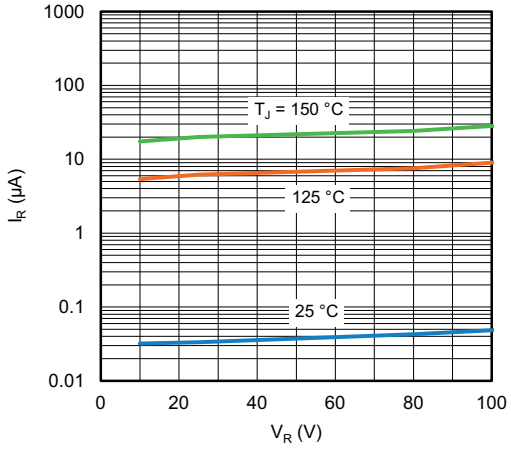



Fig. 3 - Typical Reverse Leakage Current vs. Reverse Voltage

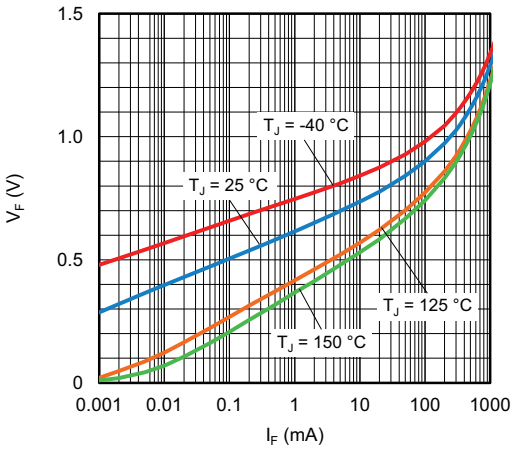


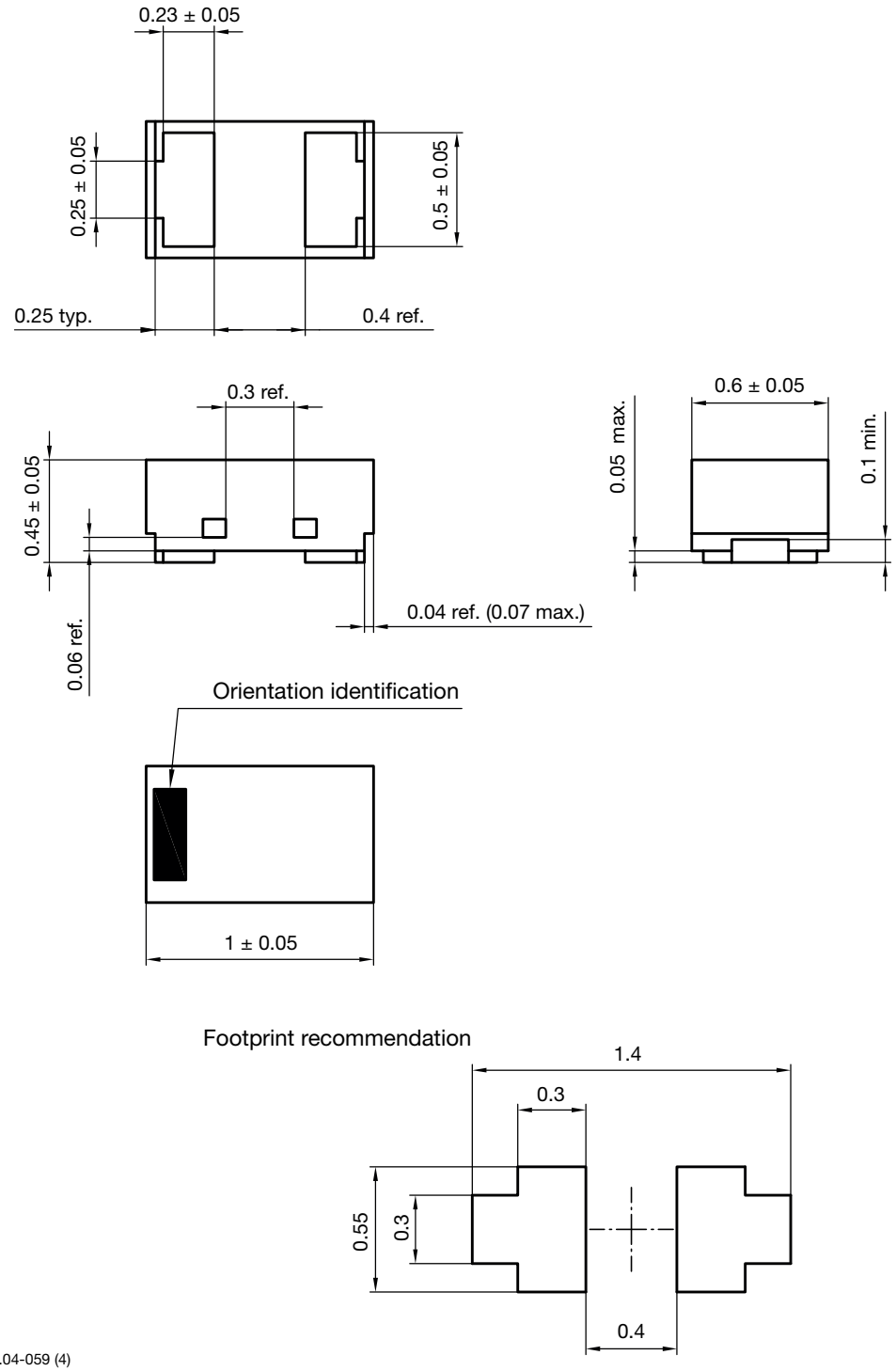

Fig. 2 - Typical Forward Voltage vs. Forward Current



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**PACKAGE DIMENSIONS** in millimeters: **DFN1006-2A**




Footprint recommendation

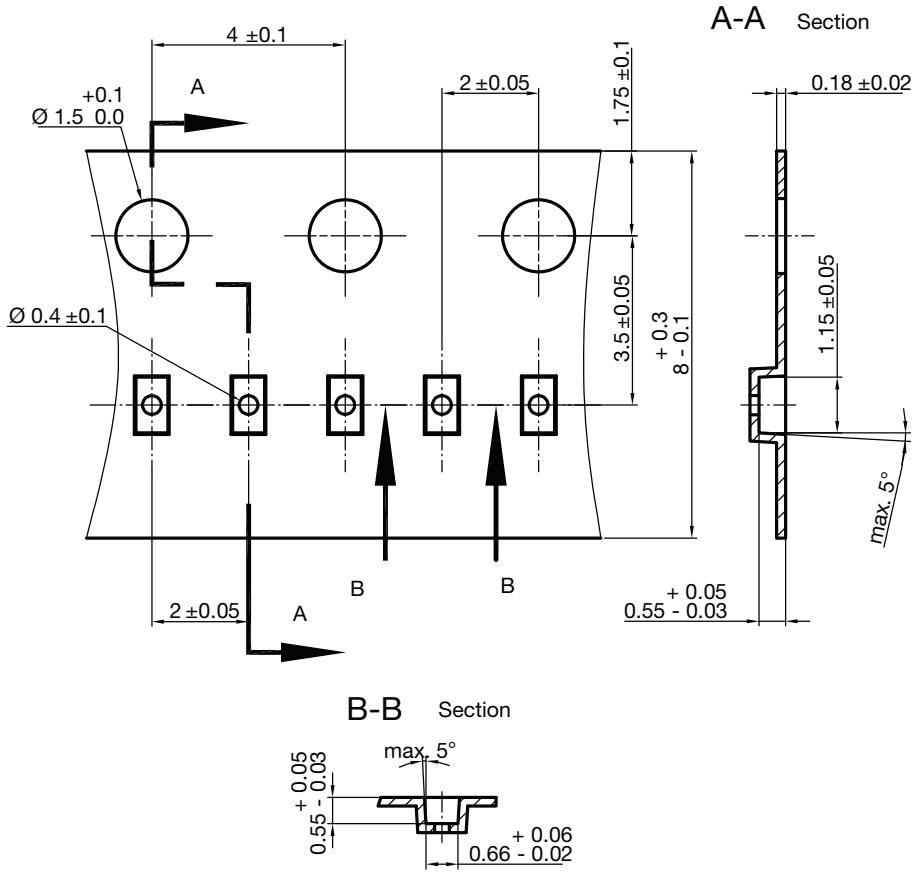



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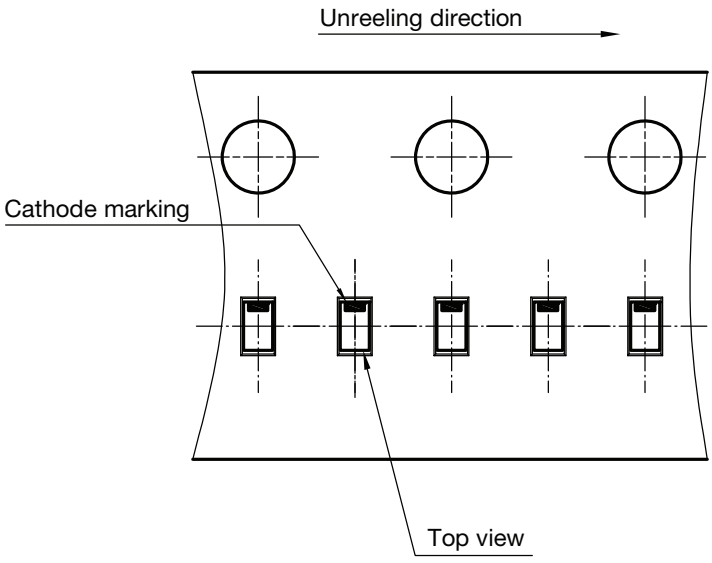
## CARRIER TAPE DFN1006-2A




S8-V-3906.04-063 (4)
created 28.10.2019

surface resistance: $10^5$ - $10^{11}$ $\frac{\text{OHMS}}{\text{SQ}}$
Cummulative tolerances of 10 sprocket holes is ± 0.2 mm

## ORIENTATION IN CARRIER TAPE DFN1006-2A




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